PHYSICS

1) The respective number of significant figures for the numbers 28.028, 0.0008, 2.4×10^{-3} are :-

(1) 4, 4 and 2
(2) 5, 1 and 2
(3) 5, 1 and 5
(4) 5, 5 and 2

2) The dimensional formula of work done is $[M^a L^b T^c]$. The value of a + b + c will be :-

(1) 5
(2) 3
(3) Zero
(4) 1

3) If $x = \frac{a\sin\theta + b\cos\theta}{\sqrt{a^2 + b^2}}$, then :-

(1) Dimension of x and a must be same.
(2) Dimension of a and b are not same.
(3) Dimension of x is same as $\sqrt{a^2 + b^2}$
(4) x is dimensionless

4) If u_1 and u_2 are units in two systems of measurement and their numerical values are 20 & 60 in two system, then :-

(1) $3u_1 = u_2$
(2) $\frac{u_1}{u_2} = 3$
(3) $u_1.u_2 = 12$
(4) $\frac{u_2}{u_1} = 1$

5) In a new system of unit, unit of mass is x kg, unit of length is y m and unit of time is z sec. The value of 100 N in this system will be :-

(1) $\frac{z^2}{100\,xy}$
(2) $100\frac{xy}{z^2}$

(3) $100\frac{z^2}{xy}$

(4) 100 xyz^2

6) If energy E, velocity v and time T are taken as new fundamental quantities, then dimensional formula of surface tension will be :-

(1) $[EV^2T^2]$

(2) $[E^{-1}V^2T^2]$

(3) $[EV^{-2}T^{-2}]$

(4) $[E^{-1}V^2T^{-2}]$

7) $[ML^2T^{-3}A^{-1}]$ is the dimensional formula for :-

(1) Electric potential

(2) Resistance

(3) Current

(4) Electric field

8) If percentage error in the measurement of mass and velocity are 1% and 2%, then percentage error in kinetic energy will be :-

(1) 3%

(2) 2%

(3) 5%

(4) 1%

9) The random error in the arithmetic mean of 100 observations is x, then random error in arithmetic mean of 600 observation would be :-

(1) 6x

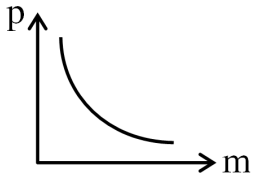
(2) $\frac{x}{6}$

(3) 3x

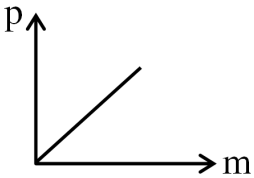
(4) x

10) If the kinetic energy of a body remains constant, then its momentum (p) - mass (m) graph is :-

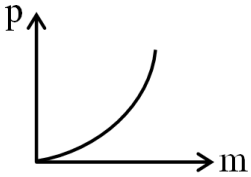
(1)



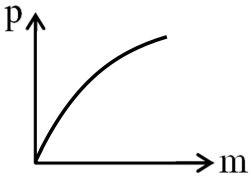
(2)



(3)



(4)



11) If a ball of mass 4 kg is dropped from a height 20 m on a floor and rebounds and travels maximum height of 15m, then total workdone by gravity for this downward and upward motion is :- (consider no air resistance and take g =10 m/s^2)

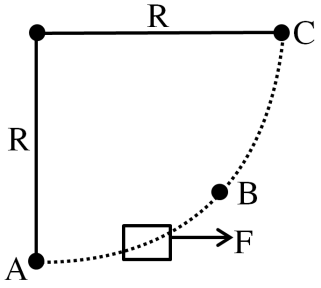
(1) 120 J

(2) 200 J

(3) 800 J

(4) –600 J

12) A block of mass m is taken from point A to point B under the action of a constant force F along a half path of a quarter circle of radius R. Workdone by this force is :-



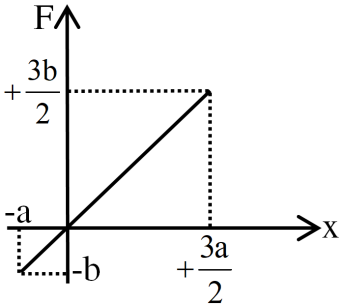
(1) FR

(2) $FR/\sqrt{2}$

(3) $\sqrt{2}\,FR$

(4) $\frac{\pi}{2}FR$

13) A force F acting on a particle varies with position x, as shown in the graph. Find the workdone by the force in displacing the particle from x = –a to $x = +\frac{3}{2}a$.



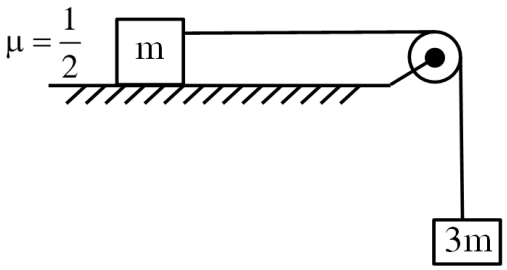
(1) ab/2

(2) $\frac{3}{2}ab$

(3) $\frac{5}{8}ab$

(4) $\frac{9}{4}ab$

14) Consider the following situation as shown in the figure. The system is released from rest. The kinetic energy of mass m, when it has travelled a distance x on rough surface, will be :- (String is inextensible and pulley is ideal)



(1) mgx

(2) $\frac{1}{2}$(mgx)

(3) $\frac{5}{8}$(mgx)

(4) $\frac{2}{3}$(mgx)

15) A particle of mas 2 kg lying on x-axis experiences a force given by law F = x(4x–3)N, where x is the x-coordinate of the particle in meters. The point on x-axis, where particle is in stable equilibrium, is :-

(1) x = 0

(2) x = 3/4

(3) Both (1) & (2)

(4) x = 4/3

16) A force of magnitude 17.32 N acting along $(\hat{i} + \hat{j} + \hat{k})$, displaces a particle from point (1, n, n) to (n, n + 1, n + 2) and workdone during this displacement is 36 J, then the value of n is :-

(1) 1.6

(2) 2

(3) 3.8

(4) 2.6

17) **Assertion (A) :** Work done in moving a body over a closed loop is zero for every force in nature.
Reason (R) : Work done does not depend on nature of force.

(1) Both (A) & (R) are correct and (R) is correct explanation of (A).

(2) Both (A) & (R) are correct but (R) is not correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) Both (A) and (R) are incorrect.

18) A force F(x) is conservative, if -
(p) It can be derived from potential energy U(x).
(q) Work done by F(x) depends only on the end points of the path.
(r) Work done by F(x) in a closed path is zero.
Which of the above statement(s) is/are correct ?

(1) Only p
(2) Both p and q
(3) Only q
(4) All p, q and r

19) The potential energy of a 0.5 kg particle free to move along x-axis is given by,
U(x) = $\left(\frac{x^2}{2} - x\right)$J
The total mechanical energy of the particle is 1 cal, then maximum kinetic energy (in J) is :-

(1) 4.7
(2) 3.6
(3) 2.8
(4) 1.5

20) A constant power P is applied to a particle of mass m. The distance travelled by the particle, will be when its velocity increases from v to 3v, will be (neglect friction)

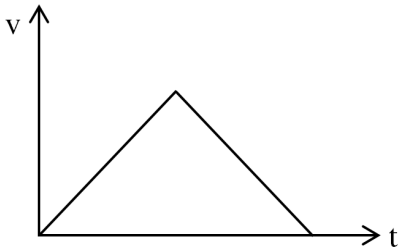
(1) $\frac{26}{3}\frac{mv^3}{P}$
(2) $8\frac{mv^3}{P}$
(3) $\frac{7}{3}\frac{mv^3}{P}$
(4) $\frac{mv^3}{P}$

21) Which of the following statement(s) is/are correct?
(a) Absolute value of potential energy cannot be determined.
(b) Change in potential energy is reference independent.
(c) Potential energy can also be defined for non conservative forces also.

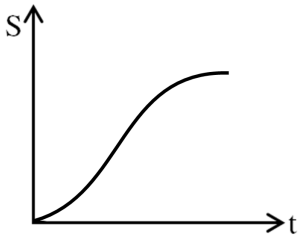
(1) Only (a) & (b)
(2) Only (b) & (c)
(3) All (a), (b) & (c)
(4) None of these

22)

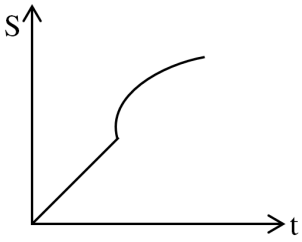
Which is the best displacement-time graph for given velocity-time graph ?



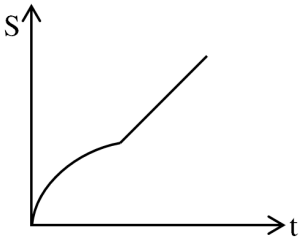
(1)



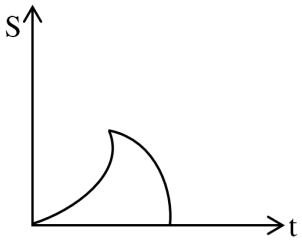
(2)



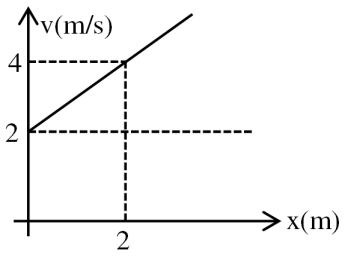
(3)



(4)



23) Find out acceleration of particle at x = 2m in given v-x graph.



(1) 2 m/s^2

(2) 4 m/s^2

(3) 1 m/s^2

(4) 3 m/s^2

24) Car A is moving towards north with speed 10m/s, its driver observes that car B is moving towards east with speed 10m/s. Find out direction of car B.

(1) N-W

(2) S-E

(3) N-E

(4) S-W

25) A man is running on level road with speed 4m/s. He feels rain drops are falling vertically downwards. If speed of rain drops is 8 m/s with respect to ground, find out speed of rain drops with respect to man.

(1) 4 m/s

(2) $4/\sqrt{3}$ m/s

(3) $4\sqrt{3}$ m/s

(4) $2\sqrt{3}$ m/s

26) A 24 m wide river is flowing with speed 3 m/s. A man can swim with speed 4 m/s in still water. He crosses the river in shortest time. Find out his displacement.

(1) 18 m

(2) 24 m

(3) 30 m

(4) 40 m

27) A partical is projected with speed 50 m/s at an angle 53° from horizontal. Find out rate of change of speed at 1s.

(1) 5 m/s^2

(2) $5\sqrt{3}$ m/s^2

(3) $5\sqrt{2}$ m/s^2

(4) 10 m/s^2

28) A particle is dropped from top of a tower. If it covers $\frac{5h}{9}$ height in last sec. of motion, then find out height of the tower.

(1) 45 m

(2) 80 m

(3) 125 m

(4) 180 m

29) If position of a particle is given by x = $(3t^2 - 5t + 4)$ m, then find out average velocity in first 3s.

(1) $\frac{16}{3}$ m/s

(2) 7 m/s

(3) 4 m/s

(4) $\frac{5}{3}$ m/s

30) A particle covers one third distance with speed 4 m/s and rest distance with speed 2 m/s. Find out its average speed in whole journey.

(1) $\frac{6}{5}$ m/s

(2) $\frac{9}{4}$ m/s

(3) $\frac{11}{3}$ m/s

(4) $\frac{12}{5}$ m/s

31) If $S_{n^{th}} = u + \frac{a}{2}[2n-1]$, then
(A) Above equation is correct in physics
(B) Above equation is dimensionally correct
(C) Above equation is dimensionally wrong
(D) Above equation is not correct in physics
Choose correct option :-

(1) A,C

(2) A,B

(3) C,D

(4) B,D

32) A particle is projected vertically upward. It covers equal distance in 4th & 9th second. Choose correct option:-

(A)	Time of flight (sec.)	(P)	180
(B)	Maximum height (m)	(Q)	60
(C)	Projection velocity (m/s)	(R)	12
		(S)	100

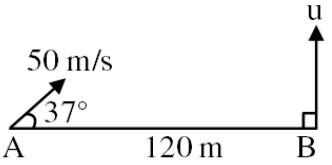
(1) (A) → P, (B) → P, (C) → S

(2) (A) → S, (B) → Q, (C) → R

(3) (A) → R, (B) → P, (C) → Q

(4) (A) → Q, (B) → R, (C) → P

33) Find out collision time of projectile A & B in given figure.



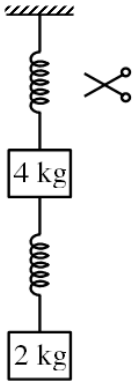
(1) 2 s

(2) 3 s

(3) 4 s

(4) 5 s

34) Find out acceleration of 2 kg body just after cutting the upper spring :



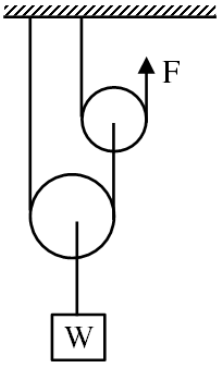
(1) g↓

(2) $\frac{g}{2}\downarrow$

(3) Zero

(4) $\frac{g}{4}\downarrow$

35) Find out mechanical advantage of given machine in diagram :



(1) 2

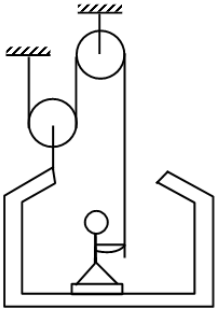
(2) 3

(3) 4

(4) 6

36) A man of 70 kg is standing on a weighing machine in a cage of 50 kg, as shown in the figure.

Find out tension in the string if system is in equilibrium :



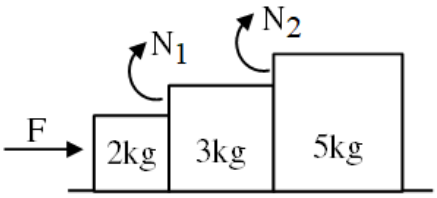
(1) 300 N

(2) 400 N

(3) 600 N

(4) 900 N

37) Find out ratio of normal contact force N_1 and N_2 :



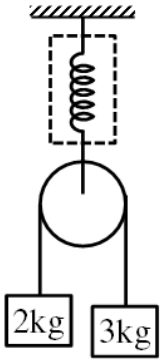
(1) $\frac{5}{3}$

(2) $\frac{8}{5}$

(3) $\frac{5}{9}$

(4) $\frac{3}{5}$

38) Find out reading of spring balance in given figure :



(1) 12 N

(2) 24 N

(3) 48 N

(4) 50 N

39) A machine gun is mounted on a 2000 kg car standing on a horizontal frictionless surface. The gun fires bullets of mass 10 gm with a velocity of 500 m/s with respect to the car. If the number of bullets fired per second is ten, the average thrust on the system is :

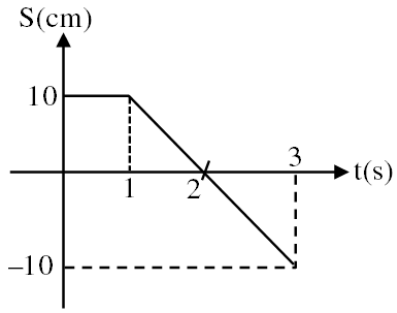
(1) 550 N

(2) 50 N

(3) 250 N

(4) 250 dyne

40) Find out impulse in interval 1s to 3s in the given displacement time graph. (m = 1kg)

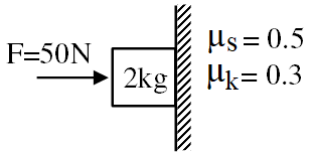


(1) 10 kg-m/s

(2) 20 kg-m/s

(3) 30 kg-m/s

(4) Zero



41) Find out friction on the block in given figure :

(1) 25 N

(2) 20 N

(3) 15 N

(4) 10 N

42) A man of mass 90 kg is standing in an elevator whose cable broke suddenly. If the elevator falls freely the force exerted by the floor on the man is :

(1) 90 N

(2) 900 N

(3) 0 N

(4) 40 N

43) With what minimum acceleration can a fireman slide down a rope whose breaking strength is $\frac{3}{4}$th of his weight ?

(1) $\frac{1}{4}g$

(2) g

(3) $\frac{3}{4}g$

(4) Zero

44) Find out True (T) and False (F) among following statements :
(A) Friction depends on weight of the body.
(B) Kinetic friction is always less than static friction.

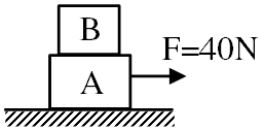
(1) TT

(2) TF

(3) FT

(4) FF

45) A block B of mass 2kg is placed on another block A of mass 3 kg which rests on a smooth horizontal surface. If μ = 0.5 between A and B and a force of 40 N is applied on A, the acceleration of A is :



(1) 4 m/s^2

(2) 8 m/s^2

(3) 10 m/s^2

(4) 12 m/s^2

CHEMISTRY

1)

The molecular formula of a compound is X_4O_9. If the compound contain 40% X by mass. Atomic mass of X is :-

(1) 24

(2) 12

(3) 26

(4) 13

2) Mole fraction of urea in its 0.5 molal aqueous solution is:

(1) $\frac{1}{112}$

(2) $\frac{2}{111}$

(3) $\frac{1}{224}$

(4) $\frac{1}{222}$

3) If 10 moles of CH_4 reacts with 21 mol O_2 at STP. The no. of moles of excess reactant that remains unreacted after the completion of reaction will be :-
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

(1) 1 mol O_2

(2) 2 mol O_2

(3) 1 mol CH_4

(4) 2 mol CH_4

4) 10 ml of uncombined O_2 is left after reaction if 4 mL of H_2 reacts with O_2 then the final volume of the mixture will be

$$H_2(g) + \frac{1}{2} O_2(g) \rightarrow H_2O(g)$$

(1) 12 mL

(2) 14 mL

(3) 8 mL

(4) 2 mL

5) When mixture of 6 L of CH_4 and Propane were burnt 7 L CO_2 was obtained then CH_4 and propane in mixture will be :-

(1) 5.5 L and 0.5 L

(2) 6 L and 1 L

(3) 3 L and 4 L

(4) 5 L and 2 L

6) Which of the following pairs of substances illustrates the law of multiple proportions?

(1) CO and CO_2

(2) NaCl and NaBr

(3) H_2O and D_2O

(4) MgO and $Mg(OH)_2$

7) Which of the following has highest mass

(1) 2 g atom of C

(2) 2 mol of CH_4

(3) 20 mL water

(4) 5 mol of O_2 gas

8) The density of a gas at STP is 2.86 g/L. Its molecular mass would be :-

(1) 16

(2) 32

(3) 48

(4) 64

9) For which reaction will $K_p = K_c$?

(1) $2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$

(2) $H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$

(3) $C(s) + 2H_2O(g) \rightleftharpoons 2H_2(g) + CO_2(g)$

(4) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

10) For the gas phase reaction;
$C_2H_{4(g)} + H_{2(g)} \rightleftharpoons C_2H_{6(g)}$;
$\Delta H = -32.7$ kcal carried out in a closed vessel, the number of equilibrium moles of C_2H_4 can be increased by :

(1) Increasing temperature

(2) Decreasing pressure

(3) Removing some amount of H_2

(4) All of these

11) For the reaction : $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$, the forward reaction at constant temperature is favoured by :-
(a) Introducing inert gas at constant pressure
(b) Introducing inert gas at constant volume
(c) Introducing PCl_5 at constant volume
(d) Introducing Cl_2 at constant volume

(1) a, b

(2) a, c

(3) b, c

(4) a, d

12)

Consider the following equilibrium in a closed container
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

At a fixed temperature, the volume of a reaction container is halved. For this change, which of the following statements holds true regarding the equilibrium constant (K_p) and degree of dissociation (α)?

(1) Neither K_P nor α changes

(2) Both K_P and α change

(3) K_P changes but α does not change

(4) K_P does not change but α changes

13) In the reaction, $A(s) + B(g) \rightleftharpoons 2C(s) + 2D(g)$ equilibrium is established. If the pressure of B is doubled then to re-establish the equilibrium, how many times of pressure of D becomes as compared to initial pressure-

(1) 2

(2) 3

(3) $\sqrt{2}$

(4) $\sqrt{3}$

14) K_c for $A + B \rightleftharpoons C + D$ is 10 at 25°C. If a container contains 1, 2, 3, 4 mol/litre of A, B, C and D respectively at 25°C, the reaction will proceed

(1) Left to right

(2) Right to left

(3) Already at equilibrium

(4) None of the above

15) The equilibrium constants for the formation of oxides of nitrogen are given as under (at 25°C):
(a) $N_2 + O_2 \rightleftharpoons 2NO$; $K = 5 \times 10^{-31}$
(b) $N_2 + 2O_2 \rightleftharpoons 2NO_2$; $K = 8 \times 10^{-31}$
(c) $2N_2 + 5O_2 \rightleftharpoons 2N_2O_5$; $K = 1 \times 10^{-34}$
Choose the correct statement :-

(1) NO_2 is the least stable oxide

(2) N_2O_5 is the most stable oxide

(3) The stability order should be : $N_2O_5 > NO_2 > NO$

(4) The stability order should be : $NO_2 > NO > N_2O_5$

16) Consider the following reaction : $CuO(s) + H_2(g) \rightleftharpoons Cu(s) + H_2O(g)$; $K_1 = 2 \times 10^{15}$ $H_2(g) + 0.5\ O_2(g) \rightleftharpoons H_2O(g)$; $K_2 = 5 \times 10^{22}$ The equilibrium constant for the reaction : $CuO(s) \rightleftharpoons Cu(s) + 0.5\ O_2(g)$ would be :

(1) 1×10^{38}

(2) 4×10^{-8}

(3) $K_1 + K_2$

(4) None

17) In a reversible reaction, two substances are in equilibrium. If the concentration of each one is doubled, the equilibrium constant will be :-

(1) Doubled

(2) Halved

(3) Constant

(4) Tripled

18) A 0.1 M solution of HF is 1% ionized. What is the K_a :-

(1) 10^{-5}

(2) 10^{-4}

(3) 3×10^{-5}

(4) 3×10^{-4}

19) Which one is the weakest monobasic acid? K_a value are given as under :-

(1) 1.8×10^{-5}

(2) 2.7×10^{-7}

(3) 1.4×10^{-4}

(4) 1.2×10^{-6}

20) The ratio of pH of 0.05M and 0.005M H_2SO_4 solution will be :-

(1) 2 : 1
(2) 1 : 2
(3) 1 : 15
(4) 1.5 : 1

21) pH of 10^{-10} N H_2SO_4 is :-

(1) 10
(2) 7
(3) < 7
(4) > 7

22) Which is correct for monobasic acid CH_3COOH solution ?

(1) $K_a = C\alpha^2$
(2) $\alpha = \frac{[H^+]}{C}$
(3) $K_a = \frac{[H^+]^2}{C}$
(4) all of the above are correct

23) If pH of solution is increased from 2 to 4, then concentration of $[H^+]$ ion would be :-

(1) increased 2 times
(2) decreased 2 times
(3) increased 100 times
(4) decreased 100 times

24) 25 mL, 0.2M $Ca(OH)_2$ is neutralised by 10 mL of 1M HCl. Then pH of resulting solution is :

(1) 9
(2) 1.84
(3) 12
(4) 7

25) What will be the pH and α (degree of hydrolysis) respectively for the salt BA of 0.1 M concentration? (Given: K_a for HA = 10^{-6} and K_b for BOH = 10^{-6})

(1) 5, 1%
(2) 7, 10%
(3) 9, 0.01%
(4) 7, 0.01%

26) If 100 mL, 0.1 N H_2SO_4 solution is mixed with 100 mL, 0.1 M $Ba(OH)_2$ solution then pH of resulting solution will be :-

(1) 1.3

(2) 12.7

(3) 7

(4) 1

27) If the bond energies of H–H, Br–Br and H–Br are 433, 192 and 364 kJ mol^{-1} respectively, $\Delta H°$ for the reaction, $H_{2(g)} + Br_{2(g)} \rightarrow 2HBr_{(g)}$ is

(1) –261 kJ

(2) +103 kJ

(3) +261 kJ

(4) –103 kJ

28) The enthalpy of formation of NH_3 is – 46 $\frac{kJ}{mol}$. The enthalpy change for the reaction :
$2NH_3 \rightarrow N_2 + 3H_2$ is ?

(1) 46 kJ/mol

(2) 92 kJ/mol

(3) – 23 kJ/mol

(4) – 92 kJ/mol

29) The enthalpy changes for two reactions are given by the equations

$$2Cr(s) + \frac{3}{2}O_2(g) \rightarrow Cr_2O_3(s), \Delta H = -1130kJ$$

$$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g), \Delta H = -110kJ$$

what is the enthalpy change (in kJ) for the reaction
$3C(s) + Cr_2O_3(s) \rightarrow 2Cr(s) + 3CO(g)$

(1) –1460 kJ

(2) 800 kJ

(3) –800 kJ

(4) –500 kJ

30) Which of the following reactions corresponds to the definition of standard enthalpy of formation ?

(1) C(diamond) + $O_2(g) \rightarrow CO_2(g)$

(2) C(graphite) + $O_2(\square) \rightarrow CO_2(g)$

(3) C(graphite) + $O_2(g) \rightarrow CO_2(g)$

(4) C(g) + $O_2(g) \rightarrow CO_2(g)$

31) Consider following reactions :-

(a) $N_2(g) + O_2(g) \rightarrow 2NO(g)$; $\Delta H = X_1$
(b) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$; $\Delta H = X_2$
(c) $2H_2O(g) + O_2(g) \rightarrow 2H_2O_2(g)$; $\Delta H = X_3$
(d) $PCl_3(g) + Cl_2(g) \rightarrow PCl_5(g)$; $\Delta H = X_4$

In which case(s), $\Delta H_f = \frac{\Delta H}{2}$

(1) a, b, c
(2) d
(3) b, c
(4) a

32) The heat of formation of CO_2 is –95 k cal/mol. The amount of carbon which on burning will evolve 1000 k cal is :-

(1) 12.63 g
(2) 17.95 g
(3) 126.3 g
(4) 179.5 g

33) Standard enthalpies of formation of O_3, CO_2, NH_3 and HI are 142.2, – 393.2, – 46.2 and + 25.9 kJ mol^{-1} respectively. The order of their increasing stabilities will be :-

(1) O_3, CO_2, NH_3, HI
(2) CO_2, NH_3, HI, O_3
(3) O_3, HI, NH_3, CO_2
(4) NH_3, HI, CO_2, O_3

34) Enthalpy of neutralization of HCl by NaOH is –55.84 kJ/mol and by NH_4OH is –51.34 kJ/mol. The enthalpy of ionization of NH_4OH is

(1) –107.18 kJ/mol
(2) 107.18 kJ/mol
(3) 4.5 kJ/mol
(4) –4.5 kJ/mol

35)

If $S + O_2 \rightarrow SO_2$; $\Delta H = -298.2$ kJ
$SO_2 + \frac{1}{2} O_2 \rightarrow SO_3$; $\Delta H = -98.7$ kJ
$SO_3 + H_2O \rightarrow H_2SO_4$; $\Delta H = -130.2$ kJ
$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$; $\Delta H = -227.3$ kJ

the heat of formation of H_2SO_4 will be :-

(1) –754.4 kJ

(2) +320.5 kJ

(3) –650.3 kJ

(4) –433.7 kJ

36) Molar enthalpy of combustion of $C_2H_{2(g)}$, $C_{(s)}$ and $H_{2(g)}$ are –1300, –394 and –286 kJ/mol respectively. The standard enthalpy of formation of $C_2H_{2(g)}$ is

(1) –226 kJ/mol

(2) 626 kJ/mol

(3) –626 kJ/mol

(4) +226 kJ/mol

37)

Find the number of moles of MnO_4^- needed to oxidise 2 mole of FeC_2O_4 in acidic medium

$$MnO_4^- + FeC_2O_4 \xrightarrow{H^+} Fe^{3+} + CO_2 + Mn^{2+}$$

(1) $\frac{6}{5}$

(2) $\frac{5}{6}$

(3) $\frac{3}{2}$

(4) $\frac{5}{2}$

38) In the balanced half reaction value of a, b, c will be :

$$Cr_2O_7^{-2} + aH^+ + be^- \rightarrow 2Cr^{+3} + cH_2O$$

(1) a = 2, b = 3, c = 7

(2) a = 6, b = 7, c = 14

(3) a = 14, b = 6, c = 7

(4) a = 7, b = 3, c = 7

39) Which of the following can act both as oxidant and reductant?

(1) NH_3

(2) MnO_2

(3) H_2SO_4

(4) All

40)

Decreasing order of oxidation number of Mn in the given compounds is

I. K_2MnO_4

II. Mn_2O_3

III. MnO_4^-

(1) III > II > I

(2) I > II > III

(3) III > I > II

(4) II > I > III

41) What will be the oxidation number of Ni in the ion $[Ni(CN)_4]^{-2}$?

(1) +2

(2) –2

(3) 0

(4) +4

42) Equivalent weight of N_2 in the change $N_2 \rightarrow NH_3$ is

(1) 28/6

(2) 28

(3) 28/2

(4) 28/3

43) In acidic medium, $KMnO_4$ oxidise iodide to Iodine. In this reaction oxidation state of Mn changes from.

(1) + 7 to + 2

(2) + 7 to + 4

(3) + 7 to + 6

(4) + 6 to + 4

44) A compound contains element X,Y,Z with oxidation no. +3, +5 and –2 respectively. What will be possible formula of compound containing above elements.

(1) XYZ

(2) $Y_2(XZ_3)_2$

(3) $X_3(YZ_4)_3$

(4) $X_3(Y_4Z)_2$

45) How many moles of $FeSO_4$, $H_2C_2O_4$ and FeC_2O_4 are oxidised separately by one mole of $KMnO_4$ in acidic medium respectively :-

(1) 5 ; 5/2 ; 5/3

(2) 2/5 ; 2 ; 5

(3) 5/3 ; 3/5 ; 2/5

(4) 3/5 ; 2 ; 5

BIOLOGY

1) Potato and brinjal are two different species that belong to which genus ?

(1) *Solanum*
(2) *Mangifera*
(3) *Triticum*
(4) *Capsicum*

2) **Assertion :** Genus may have more than one specific epithets.
Reason : Genus comprises a group of related species.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.
(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both assertion & reason are false.

3) Lion, leopard and tiger are belong to -

(1) Same genus
(2) Three different genus
(3) Two different genus
(4) Three different families

4) In a taxonomic hierarchy, genus is placed between :

(1) Kingdom and class
(2) Class and family
(3) Order and family
(4) Family and species

5) Match Column-I with Column-II and select the correct answer?

	Column-I		**Column-II**
(A)	Mango	(I)	Diptera
(B)	Wheat	(II)	*Solanum*
(C)	Brinjal	(III)	Sapindales
(D)	Housefly	(IV)	Poales

(1) (A–I), (B–II), (C–III), (D–IV)
(2) (A–III), (B–II), (C–IV), (D–I)
(3) (A–III), (B–IV), (C–II), (D–I)
(4) (A–II), (B–III), (C–I), (D–IV)

6) **Statement-I** : Each scientific name has two components.
Statement-II : The first word in a biological name represents the specific epithet while second word denotes the genus.

(1) Both statement I and II are correct

(2) Both statement I and II are incorrect

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct

7) There is a need to standarise the naming of living organisms such that a particular organism is known by the same name all over the world. This process is called :

(1) Identification

(2) Nomenclature

(3) Classification

(4) Systematics

8) Identify the incorrect statement :-

(1) Biological names are generally in Latin and printed in italics.

(2) The name of the author appears after the specific epithet in a scientific name

(3) Both the words in biological name, when handwritten, are separately underlined.

(4) In biological name first word denoting generic name starts with small letter while specific epithet starts with capital letter.

9) Binomial system of naming was given by-

(1) Carolus Linnaeus

(2) Ernst Mayr

(3) R.H. Whittaker

(4) Aristotle

10) Which of the following classification system gave equal weightage to vegetative and sexual characters ?

(1) Phylogenetic Classification

(2) Natural Classification

(3) Practical Classification

(4) Artificial Classification

11) Which of the following statements are correct for Linnaeus classification ?
I. He gave the two kingdom classification.
II. The two kingdom classification did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic and non photosynthetic organisms.
III. In two kingdom classification, *Chlamydomonas* and *Chlorella* are placed in kingdom protista.

(1) I, II and III

(2) Only I and II

(3) Only II

(4) Only I and III

12) Bacteria is a group which show most extensive :-

(1) Nutritional diversity
(2) Metabolic diversity
(3) Morphological diversity
(4) Structural diversity

13) **Statement-I** : Membranous extensions into the cytoplasm called chromatophores which contain pigments are found in all prokaryotes.
Statement-II : Pili and fimbriae are surface structures of the bacteria and play a role in motility.

(1) Both statement-I and statement-II are correct
(2) Both statement-I and statement-II are incorrect
(3) Statement-I is correct but statement-II is incorrect
(4) Statement-I is incorrect but statement-II is correct

14) Cyanobacteria also referred to as __A__. The cyanobacteria are __B__, colonial or filamentous, marine or terrestrial algae.

(1) A-Blue green algae, B-Multicellular
(2) A-Blue green algae, B-Unicellular
(3) A-Red algae, B-Unicellular
(4) A-Red algae, B-Multicellular

15) Which of the following statements are true ?
(a) Vast majority of bacteria are autotrophs.
(b) Bacteria reproduce mainly by fission.
(c) Mycoplasma are pathogenic to plants only.
(d) All cyanobacteria can fix atmospheric nitrogen.
Option :

(1) a, b and c only
(2) a, c and d only
(3) b only
(4) b, c only

16) Select the incorrect match pair from the following :

(1) Halophiles - High salt concentration
(2) Nostoc - Filamentous blue green algae
(3) Mycoplasma - Lack cell wall
(4) Mycoplasma - can not survive without oxygen.

17) Match the members of monera to their respected example and their activities and select the incorrect match :

	Members	**Example**	**Activities**

A.	Cyanobacteria	*Nostoc*	Nitrogen fixation
B.	Archaebacteria	*Lactobacillus*	Produce Alcohol
C.	Eubacteria	*Rhizobium*	Convert NO_2 into NO_3
D.	Mycoplsma	*Mycobacterium*	Disease in human

(1) A and B

(2) A, B and C

(3) A, C and D

(4) B, C and D

18) Match the column-I with column-II and select correct option :

Column-I		**Column-II**	
A	Dinoflagellates	I	Plasmodium
B	Diatoms	II	Longitudinal and transverse flagella
C	Euglenoids	III	Presence of silica in cell wall
D	Slime moulds	IV	Pellicle

(1) A – II, B – III, C – IV, D – I

(2) A – III, B – II, C – IV, D – I

(3) A – IV, B – III, C – II, D – I

(4) A – I, B – II, C – III, D – IV

19) Which of the following protist have two flagella a short and a long one ?

(1) Desmid

(2) Chrysophytes

(3) *Euglena*

(4) *Amoeba*

20) Which of the the following features is related to Diatoms ?
I - Usually found in marine environment
II - Microscopic
III - Float passively in water current
IV - Cheif producers in the oceans

(1) I, II, III only

(2) I, II, III, IV

(3) I, III, IV only

(4) II, III, IV only

21) Which of the following organisms is devoid of cell wall but their spores possess true walls ?

(1) *Euglena*
(2) Slime mould
(3) Diatoms
(4) *Gonyaulax*

22) Select the **correct** statement :

(1) Dinoflagellate are mostly fresh water organisms and photosynthetic.
(2) Dinoflagellate appear only in green and red colour.
(3) The cell wall of dinoflagellate has stiff cellulose plate on inner surface
(4) Toxins released by dinoflagellate may even kill other animals such as fishes.

23) Match the column-I (class of fungi) with column-II (example)

	Column-I (Class)		**Column-II (Example)**
A.	Basidiomycetes	i	*Rhizopus*
B.	Phycomycetes	ii	*Claviceps*
C.	Deuteromycetes	iii	*Ustilago*
D.	Ascomycetes	iv	*Trichoderma*

(1) A-i, B-ii, C-iv, D-iii
(2) A-iv, B-iii, C-i, D-ii
(3) A-iii, B-i, C-ii, D- iv
(4) A-iii, B-i, C-iv, D-ii

24) Study the following statements :
A. The fungi constitute a unique kingdom of heterotrophic organisms.
B. Fungi show a great diversity in morphology and habitat.
C. Most of the deuteromycetes are decomposer of litter and help in mineral cycle.
D. The deuteromycetes reproduce only by asexual spores known as ascospore.
Select the correct option.

(1) All are correct except 'D'
(2) All are incorrect except 'B'
(3) Only A and B are correct
(4) Only A and D are incorrect

25) Which of the following is not related with sexual reproduction in fungi ?

(1) Oospore
(2) Ascospore

(3) Conidia

(4) Basidiospore

26) **Statement-I :** Ascomycetes is commonly known as sac-fungi.
Statement-II: Ascomycetes are mostly multicellular but rarely unicellular.

(1) Both **Statement I** and **Statement II** are correct.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are incorrect.

27) Once sexual stage of member of _____ were discovered they were often moved to ascomycetes and basidiomycetes.

(1) Phycomycetes

(2) Oomycetes

(3) Zygomycetes

(4) Duteromycetes

28) Viruses that infect plants usually have :

(1) Double stranded RNA

(2) Single stranded RNA

(3) Double stranded DNA

(4) Single stranded DNA

29) The fungal partner in a Lichen is called mycobiont whereas the algal partner is called

(1) Glycobiont

(2) Algobiont

(3) Protobiont

(4) Phycobiont

30) Oogamous type of sexual reproduction is present in

(1) *Volvox*

(2) *Fucus*

(3) (1) and (2) both

(4) *Ulothrix*

31) Flagellated similar size of gametes are present in -

(1) *Ulothrix*

(2) *Spirogyra*

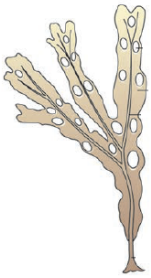
(3) *Volvox*

(4) *Fucus*

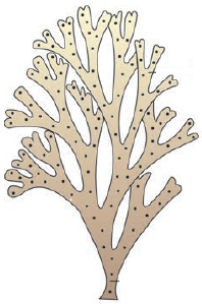
32) Identify the given diagrams (a), (b), (c) and (d)



(a)



(b)



(c)



(d)

(1) (a) *Ulothrix* (b) *Fucus*
(c) *Dictyota* (d) *Porphyra*

(2) (a) *Fucus* (b) *Ulothrix*
(c) *Porphyra* (d) *Dictyota*

(3) (a) *Ulothrix* (b) *Fucus*
(c) *Porphyra* (d) *Dictyota*

(4) (a) *Porphyra* (b) *Dictyota*
(c) *Fucus* (d) *Ulothrix*

33) Usually the members of chlorophyceae are grass green due to dominance of pigments :-

(1) Chlorophyll a & d
(2) Chlorophyll c & d
(3) Chlorophyll b & c
(4) Chlorophyll a & b

34) Which one of the following option is correct regarding stored food in the member of phaeophyceae

(1) Starch
(2) Laminarin and mannitol
(3) Floridean starch
(4) Glycogen

35) In bryophytes zygote divides and produce a multicellular body known as :-

(1) Sporophyte
(2) Gametophyte

(3) Prothallus

(4) Sporophylls

36) Choose the incorrect statement from the following :

(1) The plant body of bryophytes is more differentiated than that of algae

(2) Gemmae are green, multicellular, asexual buds, which develop in small receptacles called gemma cups.

(3) Zygote of bryophytes undergo reduction division immediately

(4) Vegetative reproduction in mosses is by fragmentation and budding in the secondary protonema.

37) Which types of Root form symbiotic association with cyanobacteria in *Cycas* ?

(1) Tap Root

(2) Fibrous Root

(3) Coralloid Root

(4) Pneumatophore

38) Which of the following statements is/are correct for *Pinus* ?

(1) Roots have fungal association in the form of mycorrhiza.

(2) Coralloid roots are associated with N_2 fixing cyanobacteria.

(3) Free-living gametophyte

(4) 2 and 3 both

39) Which of the following is an example of Sphenopsida ?

(1) *Adiantum*

(2) *Psilotum*

(3) *Equisetum*

(4) *Selaginella*

40) Match the column-I and column-II w.r.t. plant groups and their examples:-

	Column-I		**Column-II**
(i)	Chlorophyceae	(a)	*Polysiphonia*
(ii)	Phaeophyceae	(b)	*Volvox*
(iii)	Rhodophyceae	(c)	*Polytrichum*
(iv)	Moss	(d)	*Salvinia*
(v)	Heterosporous pteridophyte	(e)	*Pinus*
(vi)	Monoecious gymnosperm	(f)	*Dictyota*

(1) i-(f), ii-(a), iii-(e), iv-(c), v-(a), vi-(b)

(2) i-(b), ii-(f), iii-(d), iv-(e), v-(c), vi-(a)

(3) i-(b), ii-(f), iii-(a), iv-(d), v-(c), vi-(e)
(4) i-(b), ii-(f), iii-(a), iv-(c), v-(d), vi-(e)

41) True for bryophytes is/are -

(1) Free living gametophyte
(2) Sporophytes depends on gametophyte
(3) Both (1) and (2)
(4) Sporophyte and gametophyte both independent.

42) Which of the following pteridophyte does not produce two types of spores ?

(1) *Selaginella*
(2) *Salvinia*
(3) *Marsilea*
(4) *Lycopodium*



43) Identify the following given figure.

(1) *Funaria* a liverwort
(2) *Sphagnum* a moss
(3) *Salvinia* a pteridophyte
(4) *Equisetum* a pteridophyte



44) Which of the following option is not correct about given diagram?

(1) The main plant is heterosporous.
(2) Both seeds and fruits are produced during sexual reproduction.
(3) Pollen-tube is formed.
(4) The main plant body is sporophyte.

45) How many of the following plants belongs to Algae and Bryophyta respectively :-
Funaria, Polytrichum, Sargassum, Sphagnum, Volvox, Polysiphonia, Marchantia, Fucus, Spirogyra, Chara

(1) 4 and 6
(2) 3 and 7
(3) 6 and 4
(4) 7 and 3

46) Which one of the following features is not present in the phylum - Arthropoda ?

(1) Metameric segmentation
(2) Parapodia
(3) Jointed appendages
(4) Chitinous exoskeleton

47) In phylum porifera, the opening through, which water leaves the spongocoel is called ?

(1) Ostia
(2) Ommatidia
(3) Osculum
(4) Choanocytes

48) **Statement-I :** Platypus is an oviparous mammal.
Statement-II : Respiration in whale is done by gills.

(1) Statement I is correct.
(2) Statement II is correct.
(3) Both statement are correct.
(4) Both statement are incorrect.

49) In the reproductive system of male cockroach a characterstic……A…….is present in the 6th to 7th abdominal segments, which functions as.
Accessory reproductive gland ? Select the correct answer about 'A'

(1) Collaterial gland
(2) Mushroom gland
(3) Stink gland
(4) Salivary gland

50) Columnar epithelium with brush border is found in :-

(1) PCT of nephron
(2) DCT of nephron
(3) Lining of intestine
(4) Duct of salivary gland

51) **Assertion :** Notochord is a mesodermally derived rod like structure formed on dorsal side.
Reason : Animals with notochord are called chordates.

(1) Both assertion and reason is correct and reason is the correct explanation of assertion.
(2) Assertion is correct but reason is wrong.
(3) Both assertion and reason is correct but reason is not the correct explanation of assertion.
(4) Assertion is wrong but reason is correct.

52) Fill in the blanks and chose the correct option-
In the frog, ____A___ receives blood through the major veins called _____B____. The ventricles open into a sac-like conus arteriosus on the ____C_____ side of heart.

(1) A-Sinus arterosus, B-Venacava, C-Dorsal
(2) A-Sinus Venosus, B-Venacava, C-Ventral
(3) A-Sinus Venosus, B-Aorta, C-Ventral
(4) A-Sinus arterosus, B-Aorta, C-Dorsal

53) The name Echinodermata refers to :-

(1) An endoskeleton of calcareous ossicles.
(2) Water vascular system.
(3) Symmetry
(4) Development

54) In all connective tissues except _____ , the cells secrete fibers of structural proteins ?

(1) Bone
(2) Cartilage
(3) Adipose tissue
(4) Blood

55) Medusa is :-

(1) Sessile & cylindrical
(2) Sessile & umbrella-shaped
(3) Free living & cylindrical
(4) Free living & umbrella shaped

56) Which cell junction help to stop substances from leaking across a tissue ?

(1) Adhering junction
(2) Gap junction
(3) Tight junction
(4) Both 1 and 2

57) Choose the incorrect statement about class reptilia.

(1) body is covered by dry and cornified skin.

(2) They do not have extermal ear opening.

(3) Operculum represent the ear.

(4) They are oviparous and development is direct.

58) How many pairs of gill slits are found in cyclostomes for respiration ?

(1) 6-8

(2) 4-6

(3) 6-15

(4) 8-16

59) Match the column and select the correct option :-

	Column-I		**Column-II**
A	Cyclostomes	(i)	Lay shelled eggs
B	Amphibia	(ii)	Mammary glands
C	Aves	(iii)	External fertilization
D	Mammalia	(iv)	Without jaw

(1) A-i, B-ii, C-iv, D-iii

(2) A-ii, B-i, C-iii, D-iv

(3) A-iii, B-iv, C-ii, D-i

(4) A-iv, B-iii, C-i, D-ii

60) Which one of the following is the excretory product of Cockroaches ?

(1) Ammonia

(2) Urea

(3) Uric Acid

(4) Both (1) and (2)

61) Given below are two statements :-
Statement-I :- The head of cockroach is formed by the fusion of six segments and show great mobility.
Statement-II :- The neck of cockroach is short-extension of prothorax.
In the light of above statements choose the correct answer from the option given below.

(1) Statement I is correct but statement II is false

(2) Statement I is incorrect but II is true

(3) Both statement I and II are true

(4) Both statement i and II are false

62) In reproductive system of cockroach, the location of testes and ovary in abdomen are respectively :-

(1) 4th - 6th, 2nd - 6th

(2) 2nd - 6th, 4th - 6th

(3) 4th - 6th, 6th - 8th

(4) 6th - 8th, 8th - 10th

63) How many the following statements are correct regarding class-chondrichthyes ?
(A) It includes only fresh water fishes
(B) In males pelvic fins bears claspers
(C) They are cold-blooded (poikilothermous) animals
(D) They have internal fertilisation
(E) Gills are covered by an operculum on each side
Options

(1) Five

(2) Four

(3) Three

(4) Two

64) The Frog have ability to change the colours to hide them from their enemies, this phenomena is called :-

(1) Metamorphosis

(2) Metagensis

(3) Camouflage

(4) Metamerism

65) Which one of the following ducts open in the bidders canal of frog ?

(1) Vasa efferentia

(2) Ureters

(3) Vasa deferentia

(4) Urinogenital duct

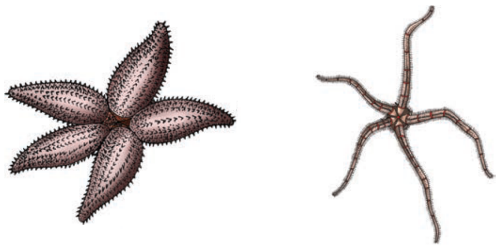
66) In birds, skin is __A__ without gland except the oilgland at the base of __B__.

(1) A-Dry, B-Tail

(2) A-Moist, B-Tail

(3) A-Dry, B-Beak

(4) A-Moist, B-Beak



(A) (B)

67) Select correct answer about given below figure A and B

(1) A-Ophiura, B-Antedon

(2) A-Cucumaria, B-Echinus

(3) A-Asterias, B-Antedon

(4) A-Asterias, B-Ophiura

68) How many of the following statements are correct regarding epithelial tissue -
A- Epithelial tissue is commonly refer as epithelium
B- The cells are compactly packed with little intercellular matrix.
C-There are two types of epithelial tissues namely simple epithelium and compound epithelium.
D-This provide a covering or a lining for some part of the body

(1) A,B

(2) A,B,C,D

(3) A,B,C

(4) B,C,D

69) Find out incorrect match :

(1) Ichthyophis - Limbless amphibia

(2) Calotes - Tree lizard

(3) Columba - Pigeon

(4) Canis - Dog

70) In echinodermates and hemichordates, the fertilisation is usually __A__ and development is __B__. Choose correct A and B respectively to complete the given statement.

(1) External, Indirect.

(2) External, Direct

(3) Internal, Indirect

(4) Internal, Direct

71) **Assertion (A)** : Compound epithelium has a major role in secretion and absorption
Reason (R) : Compound epithelium is made up of a single layer of cells and provide protection against chemical and mechanical stresses.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**

(3) **Assertion** is true but **Reason** is false

(4) Both **Assertion** and **Reason** are false.

72) Find the correct statement-

(1) Areolar tissue is a loose connective tissue
(2) Tendon is a specialized connective tissue
(3) Cartilage is a loose connective tissue
(4) Adipose tissue is a dense connective tissue

73) The moist surface of pharynx is composed of :-

(1) Compound epithelium
(2) Ciliated epithelium
(3) Cuboidal epithelium
(4) Simple squamous epithelium

74) An example of Cephalochordata is/are :-

(1) Brachiostoma
(2) Amphioxus
(3) Lancelet
(4) All of these

75) In frogs the urinary bladder present on :

(1) Dosral to stomach
(2) Ventral to rectum
(3) Ventral to stomach
(4) Dorsal to rectum

76) The epithelium of following structure provides protection against chemical and mechanical stresses :

(1) Skin
(2) Pharynx
(3) Buccal cavity
(4) All of these

77) All the cells in the epithelium lining the following structures show brush - bordered appearance except -

(1) Jejunum
(2) Fallopian tube
(3) PCT
(4) Ileum

78) What is common in Annelida, Arthropoda, Mollusca, Echinodermata and Hemichordata :-

(1) Level of organisation and symmetry
(2) Coelom and open blood circulation
(3) Tracheal respiration and reproduction
(4) Coelom and level of organisation

79) Copulatory pad in frog is present on :-

(1) First digit of forelimb of female frog.
(2) First digit of forelimb of male frog.
(3) Last digit of hind limb of male frog.
(4) First digit of hind limb of male frog.

80)

Bones have a _____ ground substance rich in calcium salts and collagen fibres-

(1) Hard and Non pliable
(2) Hard and pliable
(3) Soft and Non-pliable
(4) Soft and pliable

81) **Assertion :-** In cockroach compound eyes are situated at ventral surface of head.
Reason :- Each eye consists of about 200 hexagonal ommatidia.

(1) Both Assertion and Reason are wrong
(2) Both Assertion and Reason are true and Reason is correct explanation
(3) Both Assertion and Reason are true and Reason is not correct explanation
(4) Assertion in true but Reason is false

82) The members of phylum _____are diploblastic with two embryonic layers external ectoderm and an internal endoderm ?

(1) Platyhelminthes
(2) Aschelminthes
(3) Annelida
(4) Coelenterata

83) The haemolymph of cockroach is composed of -

(1) Colorful plasma and haemocytes.
(2) R.B.C., W.B.C and platelets.
(3) Colorless plasma and haemocytes.
(4) Colorless plasma and nephrocytes.

84) Mollusca have file like rasping organ in their mouth for feeding. It is called

(1) Ctenidia
(2) Radula
(3) Statocyst
(4) Osphradium

85) In which flatworm development is through many larval stages and fertilisation is internal ?

(1) Fasciola
(2) Ctenoplana
(3) Hookworm
(4) Limulus

86) Select the correct statement :-

(1) Annelids have open circulatory system.
(2) Hemichordates have closed circulatory system.
(3) Mollusc have water vascular system.
(4) Arthropods have open circulatory system.

87) Protochordata are :-

(1) Non-chordata
(2) Euchordata
(3) Tunicate and cephalochordates
(4) Hemichordate

88) **Statement-I :** Simple columnar epithelium is found in the lining of stomach and intestine and helps in secretion and absorption.
Statement-II : Simple squamous epithelium is made up of a single thin layer of flattened cells with irregular boundries and are not involved in functions like forming a diffusion boundry.

(1) Statement-I and II both are correct.
(2) Statement-I and II both are incorrect.
(3) Only statement-I is correct.
(4) Only statement-II is correct.

89) Read the following (a-d) four statements regarding class amphibia.
(a) As the name indicates, amphibians can live in aquatic as well as terrestrial habitats.
(b) Most of them have two pairs of limbs.
(c) Body is divisible into head, neck and trunk.
(d) Tail may be present in some.
How many as the above statements are correct ?

(1) Four
(2) Three
(3) Two

(4) One

90) In frog the cloaca is a small chamber that is used to pass :-

(1) Faecal matter
(2) Urine
(3) Gamete
(4) All of above

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	4	2	3	3	1	3	2	4	2	2	3	3	1	1	4	4	1	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	2	3	3	3	3	1	3	4	2	3	2	3	3	2	2	3	2	4
Q.	41	42	43	44	45															
A.	2	3	1	4	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	1	1	2	1	1	4	4	2	4	2	4	3	1	4	2	3	1	2	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	4	4	4	2	2	4	2	2	3	4	3	3	3	1	4	1	3	2	3
Q.	86	87	88	89	90															
A.	1	1	1	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	1	1	4	3	3	2	4	1	4	2	2	2	2	3	4	4	1	3	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	4	4	1	3	1	4	2	4	3	1	1	4	2	1	3	3	1	3	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	4	3	2	3	2	3	1	2	3	3	2	1	4	4	3	3	3	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	1	3	3	1	1	4	2	2	1	4	1	1	4	2	4	2	4	2	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	4	3	2	1	4	3	3	2	4										

SOLUTIONS

PHYSICS

1) Significant digits in 28.028 is 5, in 0.0008 is 1 and in 2.4×10^{-3} is 2.

2) Work done has dimensional formula
$[ML^2T^{-2}]$
$\Rightarrow a = 1, b = 2, c = -2$
$a + b + c = 1 + 2 - 2 = 1$

3) In $x = \dfrac{a\sin\theta + b\cos\theta}{\sqrt{a^2 + b^2}}$
$\sin\theta$, $\cos\theta$ are dimensionless.
from $\sqrt{a^2 + b^2} \Rightarrow$ a, b have same dimension.
Dimension of $\sqrt{a^2 + b^2}$ is same as a or b.
$\Rightarrow \dfrac{a}{\sqrt{a^2 + b^2}}, \dfrac{b}{\sqrt{a^2 + b^2}}$ are dimensionless.
Hence x will be dimensionless.

4) In any system of units -
nu = constant
$\Rightarrow 20u_1 = 60u_2$
$2u_1 = 6u_2$
$u_1 = 3u_2$
$\dfrac{u_1}{u_2} = 3$

5) $n_1u_1 = n_2u_2$
$\Rightarrow n_2 = n_1\left[\dfrac{u_1}{u_2}\right]$
$= 100\left[\dfrac{M_1}{M_2}\right]^1\left[\dfrac{L_1}{L_2}\right]^1\left[\dfrac{T_1}{T_2}\right]^{-2}$
$n_2 = 100\left[\dfrac{1}{x}\right]\left[\dfrac{1}{y}\right]\left[\dfrac{1}{z}\right]^{-2} = 100\dfrac{z^2}{xy}$

6) Surface tension $= \dfrac{F}{\ell} = S$
$\Rightarrow S \propto E^x V^y T^z$
$\Rightarrow [ML^0T^{-2}] = [ML^2T^{-2}]^x [LT^{-1}]^y [T]^z$
$[ML^0T^{-2}] = [M^x L^{2x+y} T^{-2x+z-y}]$
$\Rightarrow x = 1,\ 2x + y = 0,\ -2x + z - y = -2$
$y = -2, z = -2$

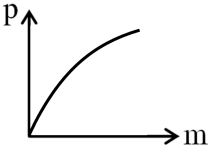
Hence, dimensional formula of surface tension will be $[EV^{-2}T^{-2}]$

7) Electric potential, $V = \frac{\text{work}}{\text{charge}} = \frac{[ML^2T^{-2}]}{[AT]}$
⇒ $[V] = [ML^2T^{-3}A^{-1}]$

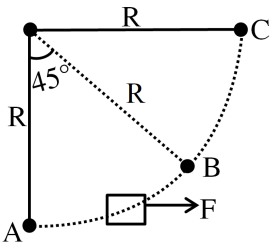
8) $K = \frac{1}{2}mv^2$
⇒ % Error in KE = % Error of m + 2 (% Error of v)
= 1% + 4% = 5%

9) If no. of observations increases n times, then random error becomes $\frac{1}{n}$ times of initial.
⇒ For 600 observations, Error = $\frac{x}{6}$

10) $p = \sqrt{2m(KE)}$
$p^2 = 2m(KE)$
(KE = constant)
comparing with $y^2 = kx$
y as p and x as m
the momentum - mass graph is parabolic



11) Total workdone by gravity during its journey :-
$w = +mgh_1 - mgh_2$
$= mg(h_1 - h_2)$
$= 4 \times 10 \times (20 - 15)$
$= 200\ J$



12)
Workdone by force F :-
W = (Force) (displacement of point of application of force in the direction of force)
W = (F)(R sin45°)
W = $FR/\sqrt{2}$

13) Workdone by force = Area under F–x graph

$$W = \frac{1}{2}\left(\frac{3}{2}a\right)\left(\frac{3}{2}b\right) - \frac{1}{2}(a)(b)$$
$$W = \frac{9ab}{8} - \frac{ab}{2}$$
$$W = \frac{5}{8}ab$$

14) $a = \dfrac{3mg - \mu mg}{(3m + m)} = \dfrac{3mg - \frac{mg}{2}}{4m}$

$$a = \frac{5g}{8}$$

for m, $v^2 = u^2 + 2as \Rightarrow v^2 = 0^2 + 2\left(\frac{5g}{8}\right)x$

$$\Rightarrow v^2 = \frac{5g}{4}x$$

$$\Rightarrow KE = \frac{1}{2}mv^2 = \frac{5}{8}(mgx)$$

15) At equilibrium position :-
F = x (4x–3) = 0
x = 0 or x = 3/4

$$\frac{dF}{dx} = 8x - 3$$

at stable equilibrium, $\dfrac{dF}{dx} < 0$

$$\left(\frac{dF}{dx}\right)_{at\,x=0} = 8(0) - 3 = -3$$
$$\left(\frac{dF}{dx}\right)_{at\,x=3/4} = 8(3/4) - 3 = 3$$

 at x = 0 → stable equilibrium.

16) $\vec{f} = f\hat{f}$

$$\vec{f} = 17.32\left(\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}\right)$$ [$\sqrt{3} = 1.732$]

$$\vec{f} = (10\hat{i} + 10\hat{j} + 10\hat{k})N$$
$$\vec{d} = (n-1)\hat{i} + \hat{j} + 2\hat{k}$$
$w = \vec{f} \cdot \vec{d}$
36 = 10 (n – 1) +10 + 20
10 (n – 1) = 6
n = 1.6

17) Workdone in the motion of a body over a closed loop is zero only when the body is moving under the action of conservative forces. It is not zero when the forces are non conservative. Hence work done depends upon nature of forces.

19) at equilibrium position :

$F = -\frac{dU}{dx} = 0$

$-(x-1) = 0$

$\Rightarrow x = 1$

$\therefore U_{min} = \frac{(1)^2}{2} - 1 = -\frac{1}{2}J$

$\therefore K_{max} = TE - U_{min}$

$= 4.2 - \left(-\frac{1}{2}\right)$

$= 4.7\ J$

20) $P = Fv = mav$

$a = \frac{P}{mv} \Rightarrow \frac{vdv}{ds} = \frac{P}{mv}$

or $v^2dv = \frac{P}{m}ds$

or $\frac{P}{m}\int_o^s ds = \int_v^{3v} v^2dv$

$\frac{P}{m}s = \left(\frac{v^3}{3}\right)_v^{3v}$

$s = \frac{m}{3P}\left(27v^3 - v^3\right)$

$s = \frac{26mv^3}{3P}$

21) Potential energy is a relative term defined only for conservative forces. Its absolute value cannot be determined.

23) $a = v\frac{dv}{dx}$, At $x = 2 \Rightarrow v = 4$ m/s

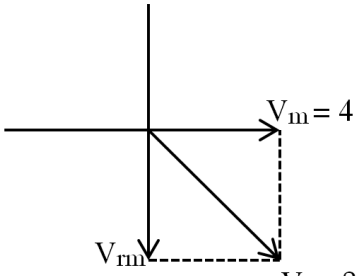
$\frac{dv}{dx} = \frac{4-2}{2-0} = 1$

$a = 4(1)$

$= 4\ m/s^2$

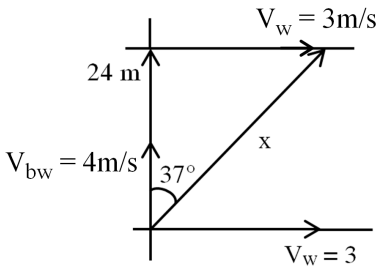
24) $\vec{v}_A = 10\hat{j} \quad \vec{v}_{BA} = 10\hat{i} \Rightarrow \vec{v}_B - \vec{v}_A = 10\hat{i}$

$\vec{v}_B = \vec{v}_{BA} + \vec{v}_A \Rightarrow \vec{v}_B = 10\hat{i} + 10\hat{j}$



25)

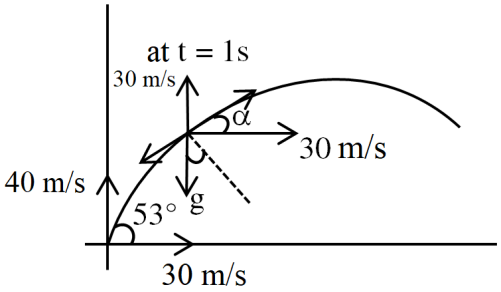
$$V_{rm} = \sqrt{V_r^2 - V_m^2}$$
$$= \sqrt{64 - 16}$$
$$= \sqrt{48}$$
$$= 4\sqrt{3} \text{ m/s}$$



26)

$$\frac{24}{x} = \cos 37^\circ$$
$$x = 30 \text{ m}$$

27) Rate of change of speed means tangential acceleration, i.e., gsinα



$$\tan\alpha = \frac{30}{30} = 1 \Rightarrow \alpha = 45^\circ$$
$$a_t = g\sin\alpha$$
$$= 10\left(\frac{1}{\sqrt{2}}\right)$$
$$= 5\sqrt{2} \text{ m/s}^2$$

28) t = 0, u = 0, a = g (const)

$s_1 : s_2 : s_3 : ..... = 1 : 3 : 5 : .....$

$$= \frac{h}{9}, \frac{3h}{9}, \frac{5h}{9}, \ldots\ldots$$

For $\frac{5h}{9}$, t = 3 s

$$h = \frac{1}{2}gt^2 = 5(3)^2$$

= 45 m

29) $t_i = 0 \Rightarrow x_i = 4$

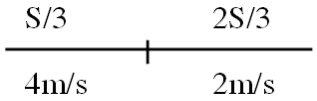
$t_f = 3s \Rightarrow x_f = 3(3)^2 - 5(3) + 4$

$= 27 - 15 + 4$

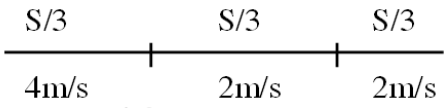
$= 12 + 4$

$<v> = \frac{x_f - x_i}{t_f - t_i} = \frac{12}{3} = 4$ m/s

30)



$\Rightarrow$



$<v> = \frac{s \times \left(\frac{3}{s}\right)}{\frac{1}{4} + \frac{1}{2} + \frac{1}{2}} = \frac{3 \times 4}{1 + 2 + 2} = \frac{12}{5}$ m/s

31)

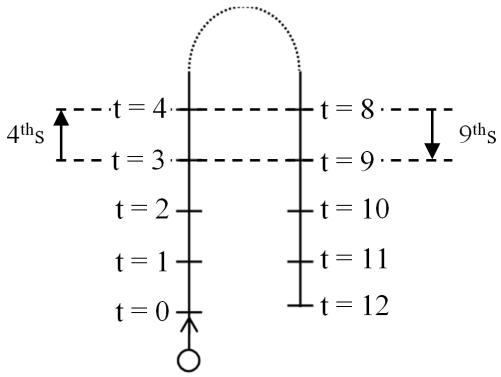
$S_{n^{th}}$ is defined for 1 sec duration,

hence $S_{n^{th}} = u + \frac{a}{2}(2n - 1)$

is specifically written as

$S_{n^{th}} = u(1) + \frac{a}{2}(2n-1)\ (1)$

Where (1) = 1 sec duration



32)

T = 12s

$T = \frac{2u_y}{g}$

$u_y = 60$ m/s

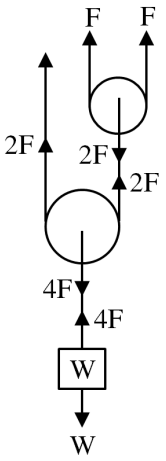
$H_{max} = \frac{u_y^2}{2g}$

$= \frac{60 \times 60}{20} = 180$ m

33) for collision
$(u_y)_A = (u_y)_B$
$u = 50 \sin 37°$
$u = 30$ m/s
along the joining line
$u_{AB} = 50 \cos 37°$
$= 40$ m/s

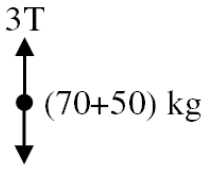
$$t = \frac{120}{u_{AB}} = \frac{120}{40}$$

$t = 3s$



35)

$$4F = W \Rightarrow \boxed{\frac{W}{F} = 4}$$



36) 1200
$3T = 1200$
$\boxed{T = 400}$

37) $N_1 = (3 + 5)\, a$
$N_2 = (5)\, a$

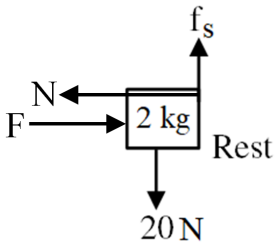
$$\frac{N_1}{N_2} = \frac{8}{5}$$

38) $T = \dfrac{2(30) + 3(20)}{5} = \dfrac{120}{5} = 24$ N
Reading $(2T) = 48$ N

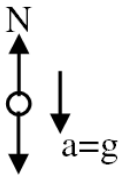
39) $F = \dfrac{dP}{dt}$
$F = nmv$

$F = 10 \times 10 \times 10^{-3} \times 500$
$F = 50$ N

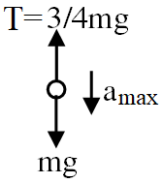
40) $\vec{I} = \Delta\vec{p} = m\Delta\vec{v}$
$\Delta\vec{v} = 0$ (v = constant)
$\vec{I} = 0$



41)
N = 50 N
$f_{\square} = \mu_s N = 0.5 \times 50 = 25$ N
$f_k = 15$ N
$\boxed{f_s = 20}$ N (block is at rest)



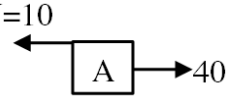
42)
N – mg = ma
$\boxed{N = 0}$



43)
$a = \dfrac{mg - \frac{3}{4}mg}{m}$
$a = \dfrac{g}{4}$

45) Block B moves due to friction only so maximum acceleration of block B is equal to
$\mu g = 0.5(10) = 5$ m/s^2

Now let assume that both blocks moves together, so their acceleration must be $\dfrac{40}{2+3} = 8$ m/s^2 , which is greater than maximum acceleration of block B so blocks will move separately.



Acceleration of A will be,

$$a = \frac{40-10}{3} = 10\,m/s^2$$

CHEMISTRY

47) $$\frac{x_B}{x_A} = \frac{mM_A}{1000}$$

49) $$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(g)}$$

4mL 2mL
(reacted)
O_2 remained = 10 mL
So, initial Vol. of O_2 = 10 + 2 = 12 mL

51) In CO : C O 12 : 16 In CO_2 12 : 32

54)

$K_p = K_c (RT)^{\Delta ng}$
If $\Delta ng = 0$ $K_p = K_c$

59) $$Q_{sp} = \frac{4 \times 3}{1 \times 2} = 6$$
$Q_{sp} < K_{eq}$
So Forward direction.

60)

Reaction having high value of K_{eq}, have more stable product.

63) $$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{0.1\left(\frac{1}{100}\right)^2}{1-0.01} \simeq 10^{-5}$$

66)

pH of Acid is less than 7.

68)

$pH_i = 2$
$[H^+]_i = 10^{-2}$ M
$pH_f = 4$

$[H^+]_f = 10^{-4}$ M

$$\frac{[H^+]_f}{[H^+]_i} = \frac{10^{-4}}{10^{-2}} = \frac{1}{100}$$

$$[H^+]_f = \frac{[H^+]_i}{100}$$

70) $$h = \sqrt{\frac{k_w}{k_a k_b}} = \sqrt{\frac{10^{-14}}{10^{-12}}} = 10^{-1}$$ ∴ % hydrolysis = $10^{-1} \times 100 = 10$

71)

$NV = (NV)_{Base} - (NV)_{acid}$

$N(100+100) = 0.2 \times 100 - 0.1 \times 100$

$$N = \frac{10}{200}$$ {Here N = OH^-}

$pOH = -\log [OH^-]$

$$pOH = -\log \frac{1}{20} = 1.3$$ pH = 14 – 1.3 = 12.7

72) $\Delta H = \Sigma(\Delta_{B.E.}H)_{Reactants} - \Sigma(\Delta_{B.E.}H)Prod.$

73) $$N_2 + \frac{3}{2}H_2 \rightarrow NH_3; \Delta H = -46\,KJ/mol$$

$$\therefore NH_3 \rightarrow N_2 + \frac{3}{2}H_2; \Delta H = +46\,KJ/mol$$

and $2NH_3 \rightarrow N_2 + 3H_2$; ΔH = +92 KJ/mol

74) eq (2) × 3 – eq(1) = eq (3)

(–110) × 3 – (–1130) = ΔH

ΔH = +800 kJ

79)

$\Delta H_{ionisation}$ = –51.34 + 55.84 = 4.5 kJ

(NH_4OH)

80) $\Delta H_f°$ SO_3 = –98.7 – 298.2

= –396.9 kJ

$\Delta rH = \Delta H_f$ $H_2SO_4 - \Delta H_f SO_3 = -130.2$

ΔH_f H_2SO_4 = –130.2 – 396.9 – 227.3

= –754.4 kJ

81) $2C_{(s)} + H_{2(g)} \rightarrow C_2H_{2(g)}$ ΔH_f = ?

$\because \Delta H = \Delta H_f = \Sigma \Delta H_{C(R)} - \Sigma \Delta H_{C(P)}$

$\therefore \Delta H_f = 2(-394) + (-286) - (-1300)$

= –788 – 286 + 1300
= +226 kJ/mol

88) $KMnO_4 + I^- \rightarrow I_2 + Mn^{+2}$

BIOLOGY

91) NCERT-XI, Pg # 7

92) NCERT-XI, Pg # 07

93) NCERT (XI) Pg # 7

94) NCERT-XI, Pg. # 07

95) NCERT-XI, Pg. # 7, 8

96)

XI NCERT Pg. No. # 6,7

97) NCERT XI Pg. # 4

98)

NCERT XI Pg.- 4, 5

99) NCERT XI Pg.- 4

100)

NCERT-XI Pg. # 24

101) XI NCERT Pg. No. # 10, 11, 12

102) NCERT-XI, Pg. No. # 13

103)

XI NCERT Pg. No. # 91

104) NCERT-XI Pg.#13

105)

NCERT-XI, Pg. # 13, 14

106)

XI NCERT Pg. No. # 13, 14

107) XI NCERT Pg. No. # 13, 14

108) NCERT-XI Pg. # 15

109)

NCERT-XI, Pg. # 15

110) NCERT XI Pg.# 14

111) NCERT-XI, Pg. # 15

112) NCERT XI Pg.# 15

113) XI NCERT Pg. No. # 17, 18

114) XI NCERT Pg. No. # 16, 18

115) NCERT-XI, Pg # 17

116) NCERT-XI Pg. # 17

117)

NCERT-XI Pg # 18

118)

XI NCERT Pg. No. # 19, 20

119) NCERT, Pg. # 21

120) NCERT-XI Pg.# 24

121) NCERT-XI Pg.# 24

122)

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123) NCERT-XI, Pg. # 26

124) NCERT-XI Pg.# 27

125) NCERT-XI Pg.# 29

126) XI NCERT Pg. No. # 29, 30

127) NCERT-XI, Pg. # 32

128) NCERT-XI, Pg.32, 33

129) NCERT (XI) Pg # 32

130)

NCERT XI - P. No. - 26-33

131)

NCERT XI Pg : 29, 30

132) NCERT-XI Page No. 32

133) NCERT-XI Page No. 31

134) NCERT-XI Page No. 33

135) NCERT -XI, Pg. # 26-30
Algae : *Sargassum, Volvox, Polysiphonia, Fucus, Spirogyra, Chara*
Bryophyta : *Funaria, Polytrichum, Sphagnum, Marchantia*

136) NCERT-XI, Pg # 44

137) NCERT-XI, Pg # 40

138) NCERT Pg # 50, 51

139) NCERT Pg. # 114, Para 7.4.2

140)

Old NCERT, Pg. # 101

141) NCERT Pg # 39

142) NCERT-XI, Pg. # 82

143) NCERT Pg. # 45

144) Module (N) Pg. # 100, Old NCERT Pg. # 102

145) NCERT, Pg. # 41

146)

Old NCERT, Pg. # 102

147) NCERT, Pg. # 49

148) NCERT Pg. # 47

149) NCERT Pg. # 46

150)

Old NCERT, Pg. # 114

151) NCERT Pg. # Module

152) NCERT Pg. # 114 (Old)

153) NCERT Pg. # 47, 48

154) NCERT-XI, Pg. # 80

155) NCERT Pg.# 83

156) NCERT Pg # 50

157) NCERT-XI Pg. # 45, Para 4.2.9

158) NCERT Pg. # 101 (Old)

159) NCERT Pg # 49

160) NCERT Pg. # 45

161) Module

162) Module

163) NCERT Pg. # 102

164) NCERT Pg. # 46

165)

NCERT Pg.# 83

166) Module

168) NCERT, Pg. # 43-45

169) NCERT Pg. # 79

170)

Module

171) NCERT Pg. # 114

172) NCERT Pg.# 38

173) NCERT Pg. # 113

174) NCERT Pg.# 44

175) NCERT, Pg # 42

176) NCERT Pg. No. # 44

177)

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178) Module

179) NCERT Pg # 48

180) NCERT-XI Pg.# 83